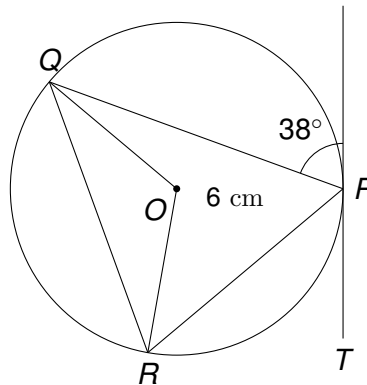


- 1 A circular lens component is being manufactured for an optical sensor. Light entering the lens travels along a straight path represented by the chord PQ , and TP is a tangent to the circle at point P .

The angle between the tangent TP and the chord PQ is 38° . The radius of the lens is 6 cm, with centre O .



PR is a diameter of the circle, so PR passes through the centre O .

- (a) By using the alternate segment theorem, find the size of angle PRQ . $PRQ = \dots\dots\dots$
(2)

- (b) Using your answer to part (a) and the fact that angle $PQR = 90^\circ$ (angle in a semicircle), calculate the length of the chord PQ , giving your answer to 3 significant figures.

$PQ = \dots\dots\dots$ cm

(2)

(Total for Question 1 is 4 marks)